Insurance Data Analysis

by

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Objectives

- Analyze trends in premiums, claims, and policy counts from 2012–2024
- Assess how pricing and claim costs affect overall profitability
- Compare new vs renewal business to understand customer behaviour
- Study the impact of economic and weather factors on insurance performance
- Identify seasonality and volatility to support future forecasting and pricing decisions

Policy Data Trend Analysis

1. Average Premium Trends

- Flat trend till 2021, sharp rise from 2022 onwards (Figure 1.1)
- Peak in 2024 — indicates premium hardening
- Likely due to inflation, claims surge, or risk repricing
- Reflects market correction after a prolonged low premium phase

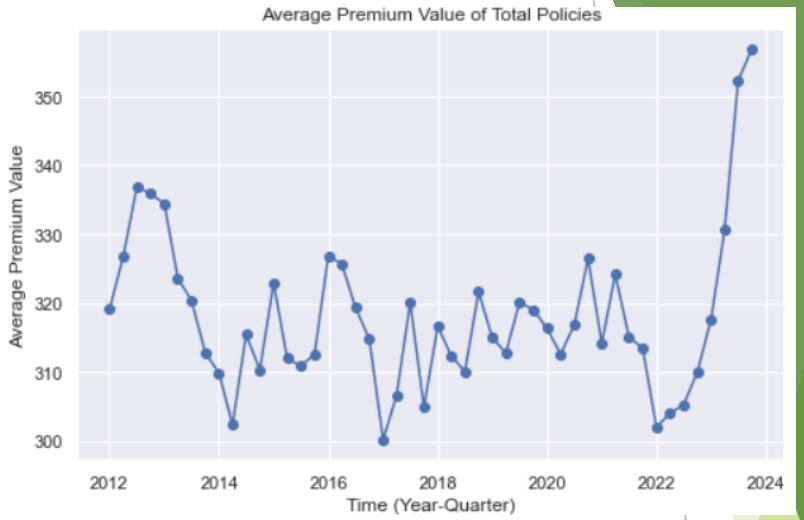

Figure 1.1- Average Premium Trends

2. Policy Volume Patterns

- Noticeable volatility with periodic peaks (Figure 1.2)
- Decline post-2021 → fewer new or renewed policies
- Higher premiums likely reduced customer demand
- Reflects affordability and retention challenges

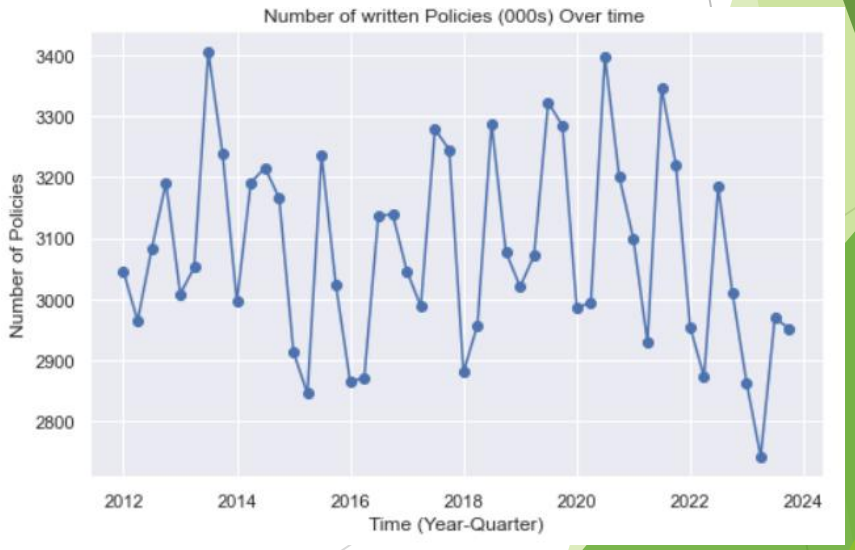

Figure 1.2- Policy Volume Patterns

3. Total Premium Value Observations

- Total premiums stable despite lower policy volumes (Figure 1.3)
- Rate increases offset reduced business volumes
- Post-2022 growth = premium hardening effect
- Suggests pricing discipline, maintaining revenue

Figure 1.3- Total Premium Value Observations

Claim Data Trend Analysis

1. Average Claim Trend

- Strong upward trend in claim cost per policy
- 2022 spike → major loss events (e.g., storm/flood impact)
- Average claim costs remain high post-2022
- Indicates rising claim severity across the market

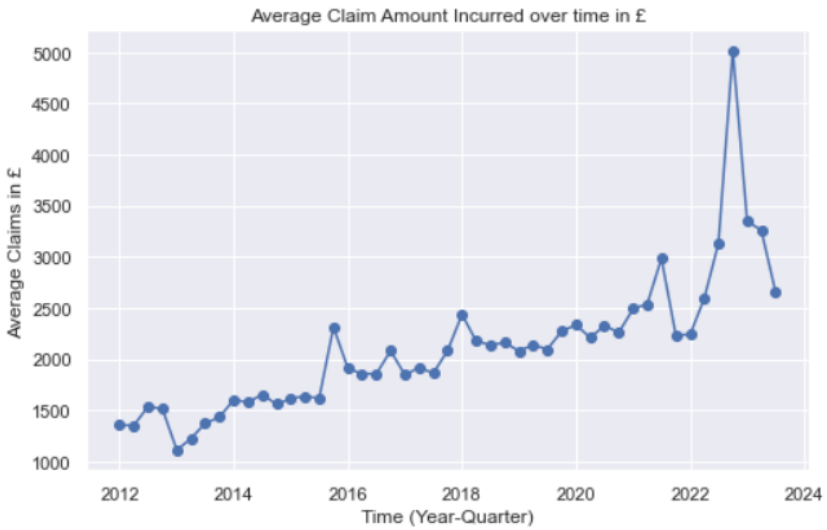

Figure 2.1- Average Claim Trend

2. Claim Volume Trend

- Sharp fall in claim counts (600k → <250k)
- Indicates fewer but larger claims
- Possible reasons: higher policy excess, better risk control
- Suggests insurers focusing on claim efficiency

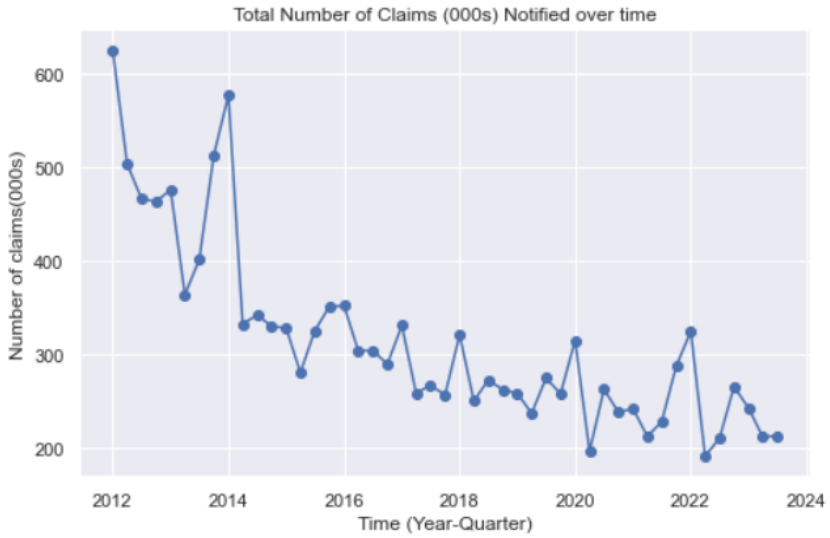

Figure 2.2- Claim Volume Trend

3. Total Claim Trend

- Volatile trend with major 2022 spike
- Lower claim frequency offset by higher claim severity
- Total cost peaks during large-scale weather events
- Reflects market exposure to catastrophe losses

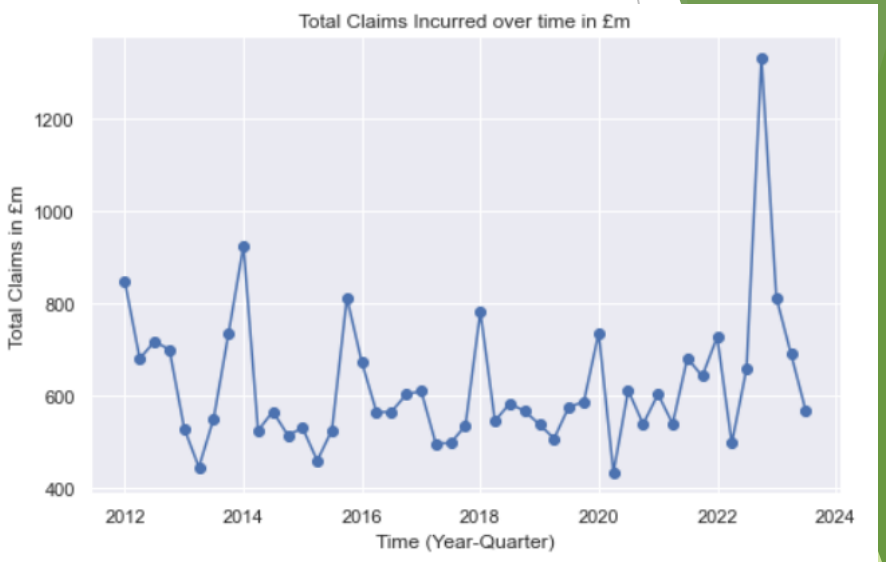

Figure 2.3- Total Claim Trend

Reasons Underlying Spikes in Number of Claims

1. Identifying Outliers in Total Number of Claims

- Before analyzing perils, outlier points were identified using boxplot analysis. Refer Figure 3.1
- Time series → ordinary boxplot sufficient (short timeframe & visual inspection)
- 7 abnormal observations detected (highlighted in Figures 3.1 & 3.2)
- These represent periods of unusually high claim volumes
- Linked to specific perils such as Storm, Flood, and Pipes+ events
- Outliers: Values > 400,000 claims

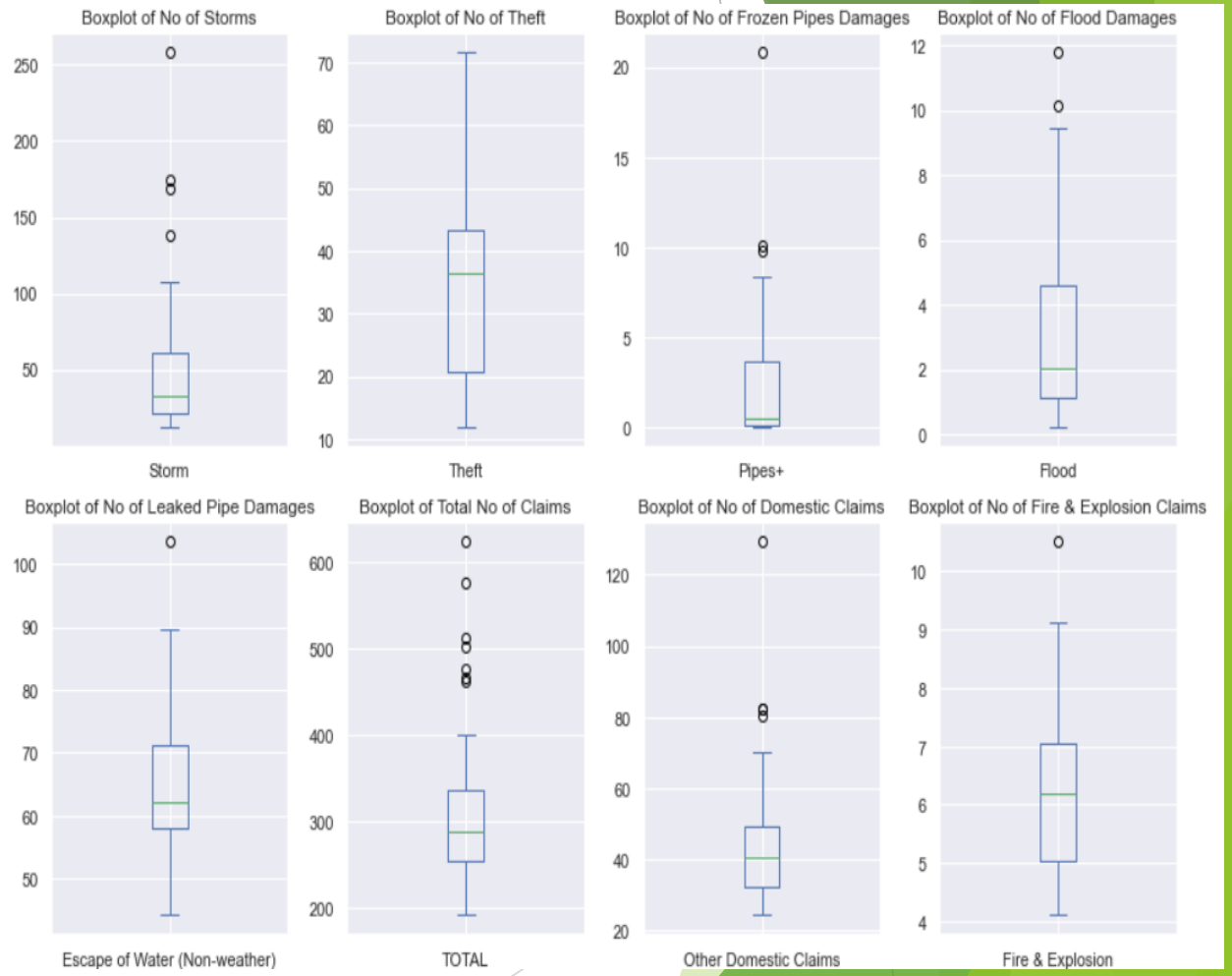

Figure 3.1- Boxplot showing outliers of claim cause and total number of claims

Year	Quarter	Fire & Explosion	Theft	Pipes+	Storm	Flood	Escape of Water (Non-weathe	Domestic Subsidenc e	Accidental Damage	Other Domestic Claims	TOTAL
2012	Q1	10.520	71.589	8.368	174.310	0.625	103.700	7.693	165.836	82.278	624.918
2012	Q2	8.885	63.100	5.848	83.126	10.152	84.899	5.807	159.152	82.364	503.333
2012	Q3	7.456	62.365	5.279	53.876	8.340	89.680	5.229	153.747	80.339	466.312
2012	Q4	8.541	68.614	4.329	67.944	7.943	84.242	4.501	146.901	70.191	463.206
2013	Q1	8.657	57.935	5.404	61.612	1.376	78.659	4.548	128.396	129.494	476.080
2013	Q4	7.457	54.721	3.851	169.291	5.716	78.759	5.159	123.546	64.512	513.012
2014	Q1	7.378	51.797	0.700	258.122	7.380	73.783	5.089	116.451	56.258	576.958

Figure 3.2- showing outlier values of Number of Claims (000s)

2. Key Outlier Periods and Causes

Table 3.3 - shows the time period, claim causes and the environmental factors linked with abnormal spikes in number of claims

Time Period	Claim Cause	Reason
2012 Q1	Storm, Pipes+, Fire & Explosion, Other Domestic	Cyclone Andrea + winter freeze → property & pipe damage
2012 Q2	Flood, Other Domestic	Extreme rainfall → widespread flooding
2012 Q3 to 2013 Q1	Other Domestic	Mild winter → mostly routine household incidents
2013 Q4 to 2014 Q1	Storm	Cyclone Dirk + 2013–14 winter storm series

3. Environmental Reasons Behind Outlier Periods

- **2012 Q1:** Cyclone Andrea → wind, rain, frozen pipes, heating failures
- **2012 Q2:** Record rainfall → river & surface flooding
- **2012 Q3–2013 Q1:** Mild winter → no major events
- **2013 Q4–2014 Q1:** Stormiest UK winter in two decades → severe wind/flood damage
- **2022 Q1 :** Storm Malik, Corrie & Dudley → regional storm damage (normal range)
- **Sources:** Met Office (2015); Kendon et al. (2015); Met Office (2022); Environment Agency (2014)

References:
*Met Office (2015) *State of the UK Climate 2014.* Exeter: Met Office.*
<https://www.metoffice.gov.uk/binaries/content/assets/metofficegovuk/pdf/weather/learn-about/uk-past-events/state-of-uk-climate/state-of-the-uk-climate-2014-v3.pdf>

*Kendon, M., McCarthy, M. & Jevrejeva, S. (2015) *The Recent Storms and Floods in the UK.* Exeter: Met Office.*
<https://www.researchgate.net/publication/310018373_The_Recent_Storms_and_Floods_in_the_UK>

*Met Office (2022) *Past UK Weather Events – Storm Malik, Corrie and December 2022 Cold Spell.* [online] Met Office.*
<https://weather.metoffice.gov.uk/learn-about/past-uk-weather-events>

*Environment Agency (2014) *The Winter Storms 2013 to 2014: UK Flooding Impact Assessment.* [online] UK Government.*
<https://www.gov.uk/government/publications/winter-floods-2013-to-2014-impacts-assessment>

4. Claim Volume Inisghts

- Long-term downward trend → better risk control & underwriting discipline
- Peak: Early 2012 (~600 claims per 1,000 policies)
- 2023–2024: <100000 claims
- Spikes: Late 2013–early 2014 (Cyclone Dirk storms) & Jan 2022 (Malik/Corrie/Dudley)
- Reflects improved resilience but continuing weather-related vulnerability

Reasons Underlying rise in Claim Amount

1) Identifying Outliers in Total Number of Claims

- *Figure 4.1*: Boxplot highlights outliers above £850m in total claim amount
- Indicates abnormally high claim quarters compared to overall distribution
- Outlier values also detected across other peril categories
- **Table 4.1**: Abnormal observations (in red) clearly visualized
- Key perils influencing spikes highlighted in yellow
- Suggests that certain peril groups (e.g., Storm, Pipes+, Flood) drive extreme claim costs
- Highlights the need to monitor high-impact perils and model extreme loss events

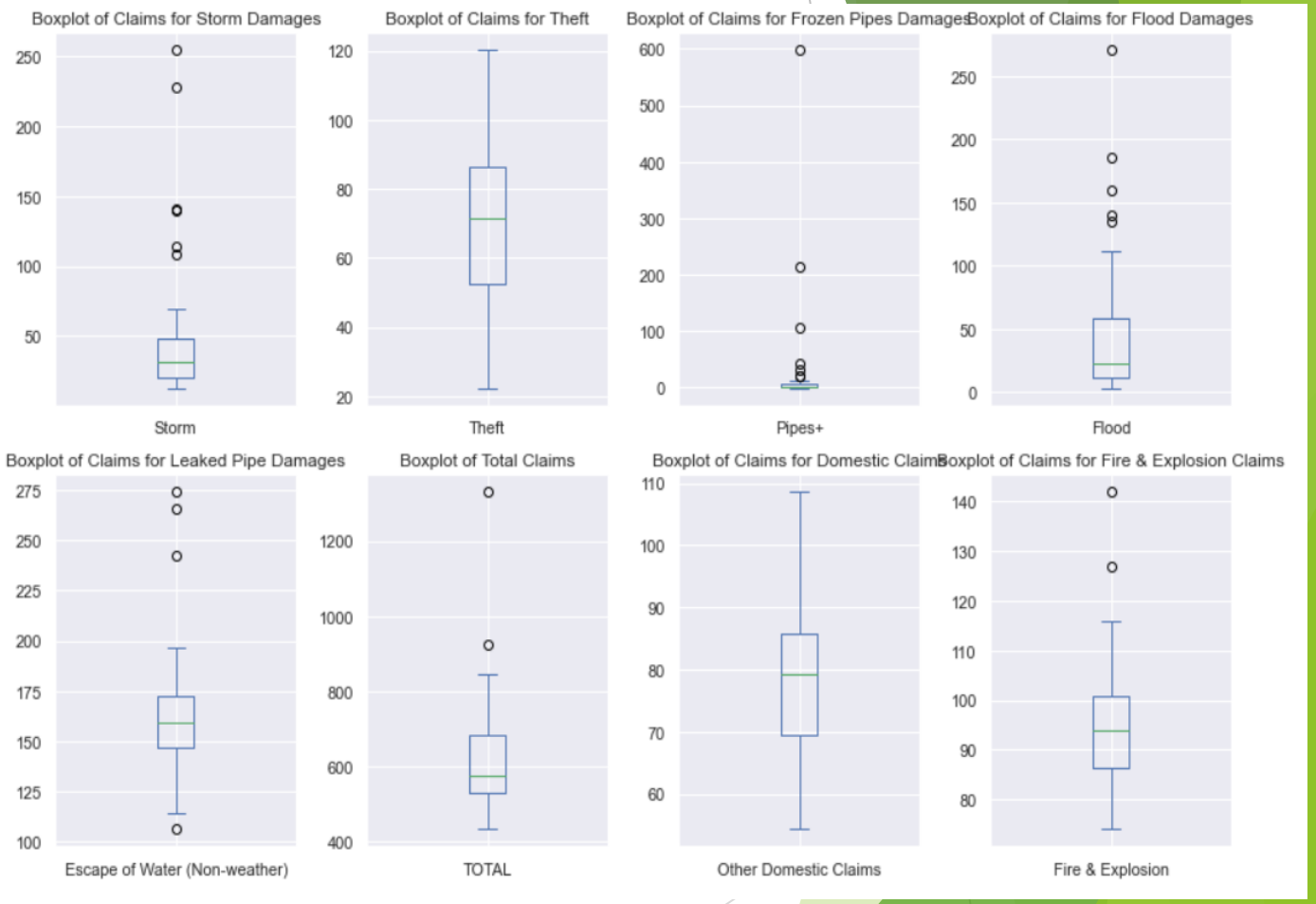

Figure 4.1- Boxplot showing outliers of claim cause and total claims amount

Table 4.1 - Abnormal spikes in claim amout

Year	Quarter	Fire & Explosion	Theft	Pipes+	Storm	Flood	Escape of Water (Non-weathe	Domestic Subsidence	Accidental Damage	Other Domestic Claims	TOTAL
2014	Q1	£79.464	£92.370	£1.738	£254.351	£185.263	£123.589	£28.724	£75.924	£83.109	£924.531
2022	Q4	£109.456	£51.977	£597.639	£44.168	£19.969	£274.247	£16.957	£69.054	£85.775	£1,331.635

2) Key Periods & Environmental Reasons

Table 4.2 - shows the time period, claim causes and the environmental factors linked with abnormal spikes in total claim amount

Time Period	Claim Cause	Reason
2014 Q1	Storm	Cyclone Dirk + 2013–14 winter storm series
2022 Q4	Pipes+, Escape of Water	Damage caused to the property through frozen & burst pipes, leakage in pipes due to the deep freeze event in Q4 2022, specifically from 8 to 18 December

3) Insights from the Claim Amount Data

- Seasonal variation seen in total claims incurred
- Two major weather-related impacts: 2014 Q1 & 2022 Q4
- Q1 2014: *Cyclone Dirk* & 2013–14 winter storms → sharp rise in claims
- Q4 2022: Claims > £1.2bn due to *frozen/burst pipes (Pipes+)*
- Deep freeze (Dec 2022): Temps below −10°C → widespread water damage
- High energy costs → reduced heating → more pipe bursts
- Zurich Insurance: Burst pipe claims ↑ ~75% YoY
- Shows extreme weather + customer behaviour raised claim severity
- Not all number of claim outliers caused large financial impact
- Focus on 2014 Q1 & 2022 Q4, where claim incurred rose
- Other outliers → small claim sizes, lower losses (backed by average claim)

Table 4.3 -
Highlighting outliner values in 'Number of Claim' that significantly increased the 'Total Claim Incurred' (marked in red), along with the specific peril catagories responsible for those spikes(marked in yellow)

Claims Incurred (£m) by Claim Cause

Year	Quarter	Fire & Explosion	Theft	Pipes+	Storm	Flood	Escape of Water (Non-weathe	Domestic Subsidence	Accidental Damage	Other Domestic Claims	TOTAL
2014	Q1	£79.464	£92.370	£1.738	£254.351	£185.263	£123.589	£28.724	£75.924	£83.109	£924.531
2022	Q4	£109.456	£51.977	£597.639	£44.168	£19.969	£274.247	£16.957	£69.054	£85.775	£1,331.635

Number of Claims Notified (000s) by Claim Cause

Year	Quarter	Fire & Explosion	Theft	Pipes+	Storm	Flood	Escape of Water (Non-weather)	Domestic Subsidence	Accidental Damage	Other Domestic Claims	TOTAL
2014	Q1	7.378	51.797	0.700	258.122	7.380	73.783	5.089	116.451	56.258	576.958
2022	Q4	4.247	20.929	10.102	40.464	1.574	69.506	7.119	79.955	31.652	265.547

Claims by Peril Over Time

The stacked area chart(*Figure 5.1*) gives the following insights:

- Other Domestic Claims shows consistent presence across all quarters.
- Its steady contribution creates a large area visually, even without extreme peaks.
- Storm and Escape of Water cause sharp spikes in certain quarters (e.g., Q1 2014, Q4 2022).
- These spikes coincide with severe weather events and frozen-pipe incidents.
- Accidental Damage and Subsidence fluctuate moderately but don't drive major surges.
- Visual dominance ≠ impact dominance: long-term steadiness vs. short-term volatility.
- Key takeaway: Combine visuals with data — consistent perils sustain baseline risk, while weather perils trigger financial peaks.

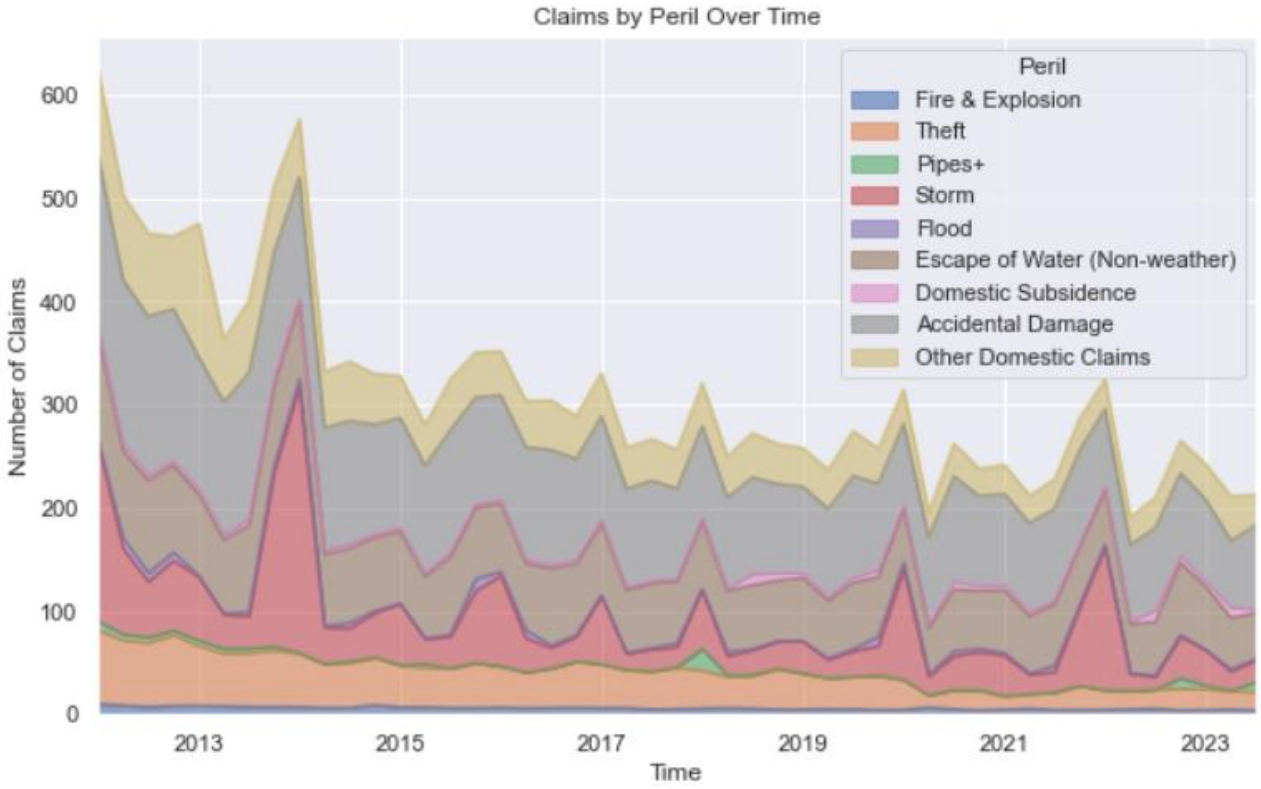

Figure 5.1 - stacked area chart of claim by peril overtime

Seasonal and Trend Analysis of Claim Amount

- Major spikes in 2014 Q1 and 2022 Q4 → driven by severe weather & frozen pipes.
- Clear seasonal pattern repeating annually — winter quarters show higher claims.
- Downward trend (2012–2018) followed by a rising trend post-2020.
- Residuals mostly stable, suggesting limited random volatility outside major events.
- Reinforces the need for weather-sensitive pricing and loss forecasting.

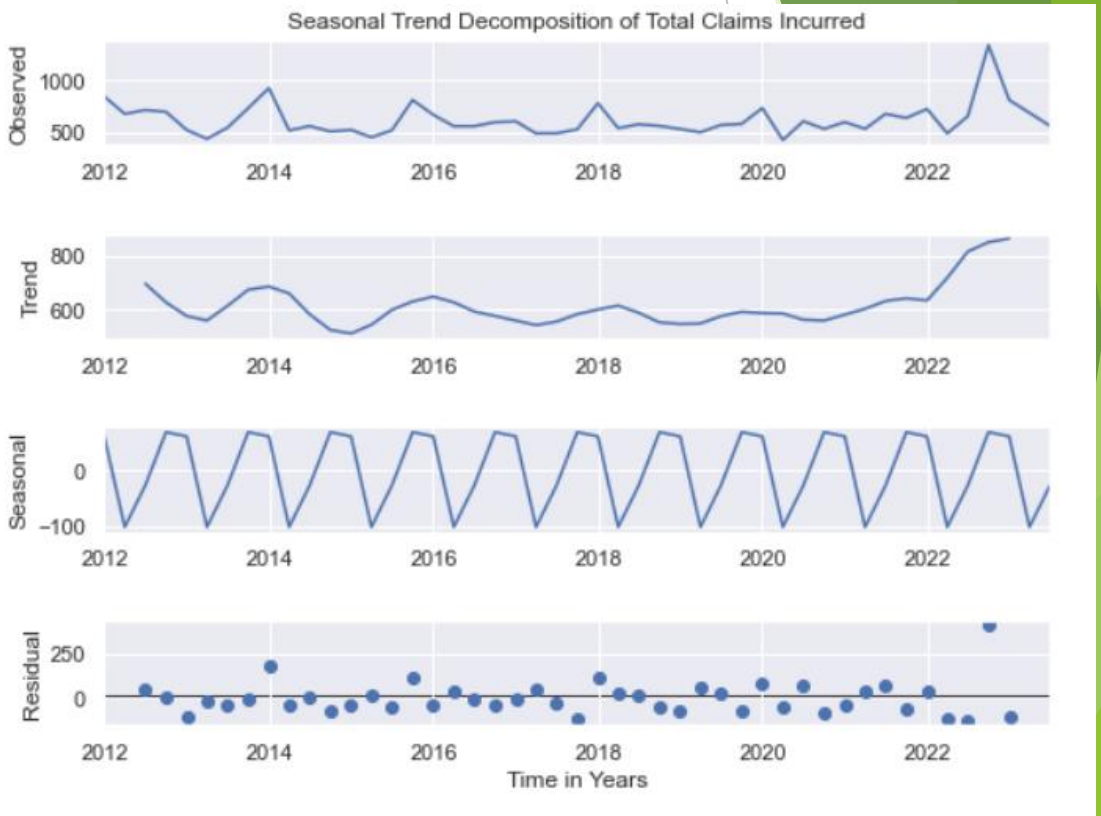

Figure 6.1- Seasonal trend decomposition of total claims incurred

Seasonal and Trend Analysis of Claim Amount

Premium vs Claims

- Premiums stable; claims show stronger fluctuations.
- Sharp claim spike in **late 2022** → weather & cost inflation.
- Premiums > claims overall → underwriting profitability.
- Narrow gaps → temporary loss ratio pressure.

Policies vs Claims

- Policy count steady (~2.8–3.3M).
- Gradual decline in claim counts since 2012.
- Low claim frequency relative to portfolio size.

Figure 7.1- Premium vs Claims

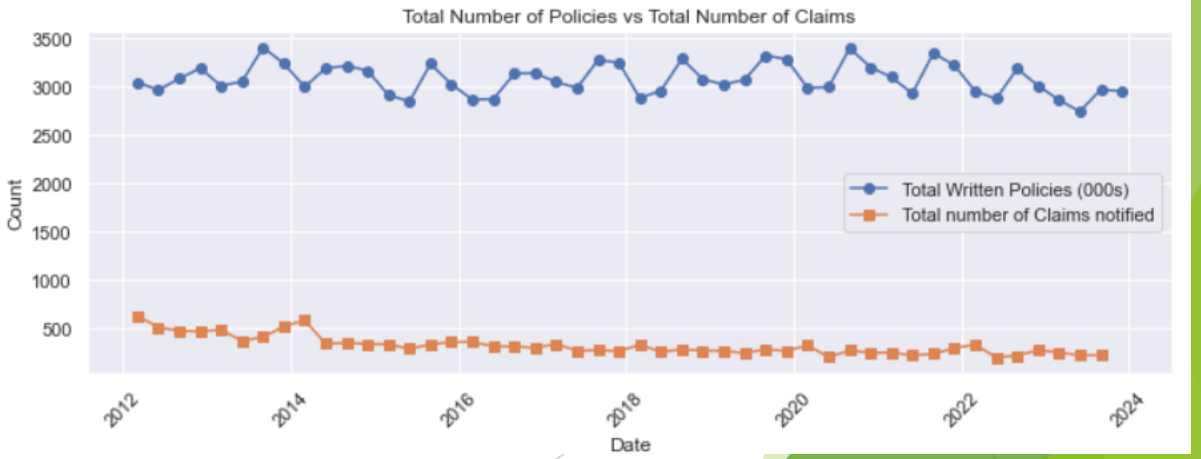

Figure 7.2- Policies vs Claims

Key Performance Metrics

1. Loss Ratio (Claims/Premium)

- Mostly 0.4–0.8 across 2012–2022 → generally profitable.
- One major spike in 2023 (~1.4) → clear loss-making period.
- Only minor local peaks around 2014, 2016, 2018, and 2021 (all ≤ ~0.9), not true spikes.

Figure 8.1- Loss Ratio

2. Average Premium

- Shows how much customers pay, on average, for insurance.
- Higher values → insurer charging more, possibly due to risk or inflation.
  Lower values → competitive pricing or softer market conditions.
- Premiums remained fairly steady between £310–£330 until 2022, but rose sharply to £350+ by 2024.
- This reflects insurers adjusting for inflation, rising construction costs, and extreme weather losses, supported by UK market data showing home insurance prices up 13–19% since 2023.

Figure 8.2- Average Premium pre Policy

3. Burn Cost (Claims per Policy)

- Represents the average claims cost per policy sold — combining both frequency and severity.
- Higher burn cost → greater claims burden per customer.
- The trend was steady for most years but peaked sharply in 2022 Q4, signalling heavier payouts per policy.
- This aligns with increased storm and freeze-related damage and rising repair expenses due to inflation

Figure 8.3- Burn Cost

4. Claim Severity (Cost per Claim)

- Shows the average payout amount per individual claim

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- Rising severity → each claim becoming more expensive to settle.
- Severity climbed gradually over time and peaked around 2022–2023, driven by higher rebuilding and material costs.
- Although it declined slightly afterwards, it remains well above pre-2020 levels — suggesting structural cost inflation in claims.

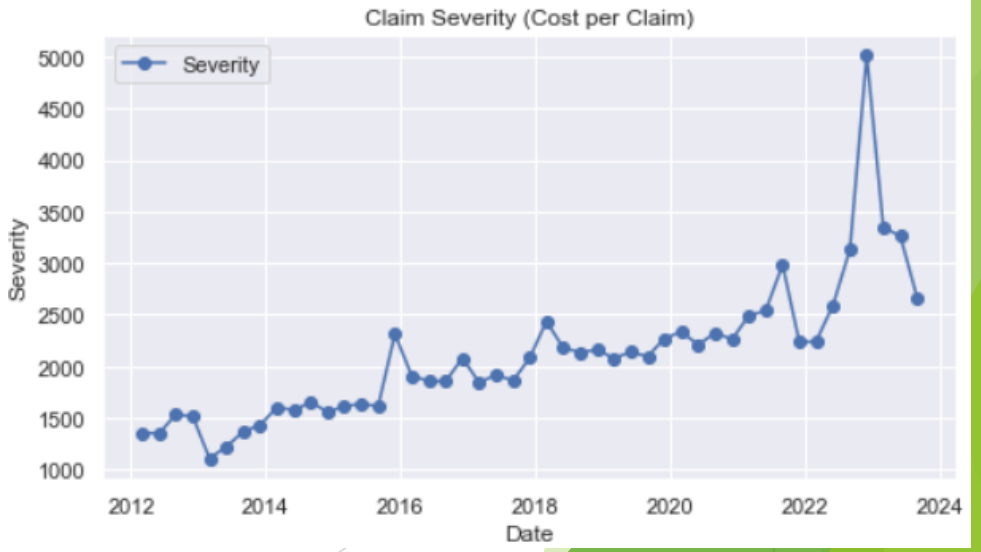

Figure 8.4- Claim Severity

5. Claim Frequency (Claims / Policy Count)

- Measures how often claims are made per policy.
- High frequency → more claims activity.
- Low frequency → improved risk management or fewer incidents.
- There's a clear downward trend from ~0.20 in 2012 to ~0.08 in 2024, showing that customers are claiming less frequently.
- This could reflect better risk prevention, safer home environments, or tighter claim validation by insurers.

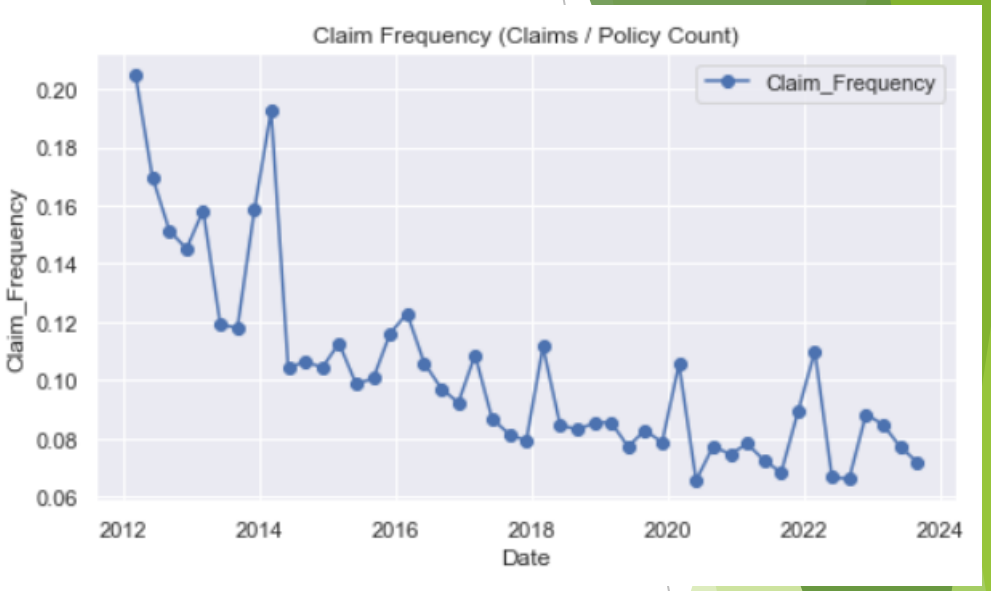

Figure 8.5- Claim Frequency

Overall Summary

- Over the 2012–2024 period, claim frequency declined significantly, but severity and burn cost rose, meaning fewer but more expensive claims.
- The loss ratio spiked during 2022–2023, coinciding with extreme weather and inflationary pressures, prompting insurers to raise average premiums to maintain profitability.
- This marks a shift from high-frequency, low-cost claims to low-frequency, high-cost risk exposure — a key factor shaping UK home insurance pricing strategies post-2022.

Profitability Check

- The comparison of Burn Cost and Average Premium indicates whether pricing adequately covers the actual risk.

- Comparing both helps assess profitability:

  If Burn Cost < Avg Premium → Profitable
  If Burn Cost ≈ Avg Premium → Break-even
  If Burn Cost > Avg Premium → Loss-making

- From the line chart, there is a clear spike in Burn Cost during Q4 2022, where it exceeded the Average Premium.

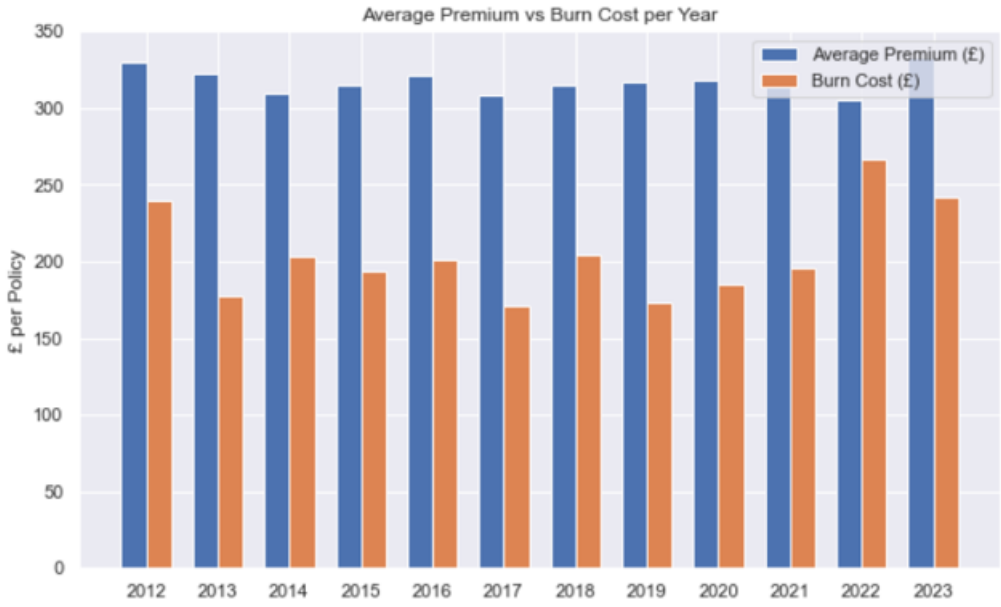

Figure 9.2– Comparison of yearly Average Premium and Burn cost

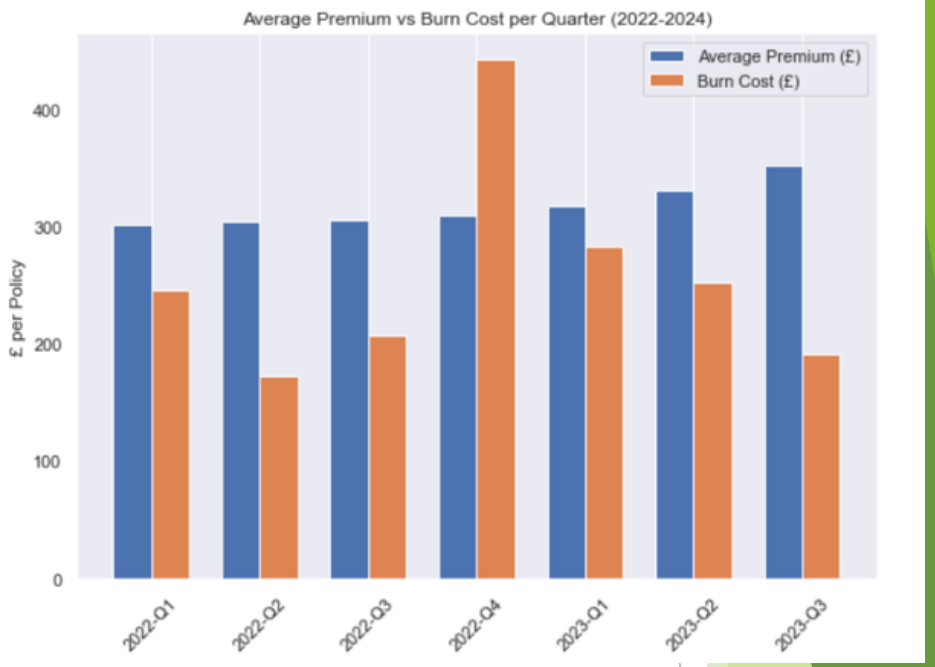

Figure 9.1– Comparison of Quarterly Average Premium and Burn cost

- This spike was driven by the December 2022 deep freeze, which led to a surge in frozen and burst pipe claims.

- The bar chart (Figure 9.1) highlights that Q4 2022 was the only quarter where Burn Cost surpassed Average Premium.

- However, the yearly comparison (Figure 9.2) shows that the Average Premium remained higher overall, indicating sustained profitability.

- This reflects effective risk management and underwriting, with insurers adjusting pricing to maintain profitability even during severe weather-related events.

Macro and Market Trends Influencing Performance Metrics

1. Premium Trends

- Flat at £310–£330 (2012–2021), sharp rise post-2022 (>£350).
- Driven by FCA pricing reform, inflation, and climate-related repricing.
- Reflects insurers adjusting prices to cover rising rebuild costs.

2. Claims Trends

- Claim Frequency ↓ — better home security, fewer thefts, cautious claiming.
- Claim Severity ↑ — inflation, supply-chain issues, severe weather (2021–2023).
- Fewer but more expensive claims overall

3. Profitability Indicators

- Loss Ratio & Burn Cost spiked in 2023 (~1.4) due to inflation and storms.
- Premium hikes followed, restoring profitability in later periods.

4. Variability & Risk Control

- High volatility from weather shocks & inflation cycles.
- Insurers use reinsurance and catastrophe models to manage risk.

Analysis of New Business Data and Renewal for Premium

1. **Customer Base Analysis**

- *Figures 11.1 and 11.2* show trend and seasonal decomposition for both.
- Analysis of New Business and Renewal Policy data to understand changes in customer base.

Insights – Renewal Policies (*Figure 11.1*)

- Overall trend stable until mid-2022, then declines through 2023.
- Clear yearly seasonal peaks, likely linked to renewal cycles.
- Small residuals indicate stable and predictable renewal patterns.
- Mild decline suggests customers are more price-sensitive or switching providers.

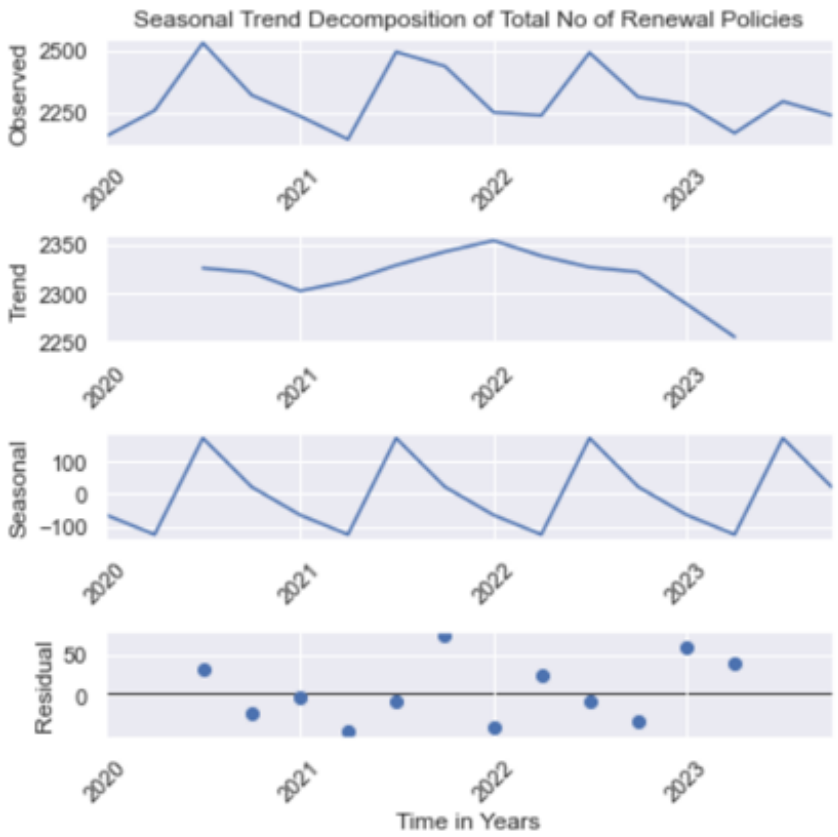

Figure 11.1- Renewal data decomposition

Insights from New Policy Data (*Figure 11.2*)

- Steady decline in trend from 2021 to 2023, showing weaker new policy acquisition.
- Regular seasonal pattern with yearly peaks and dips.
- Residuals are small and random, meaning variation is well explained by the model.
- Drop in trend reflects reduced new customer growth due to higher costs and inflation.

Business Interpretation

- Analysis of new and renewal policy volumes highlights overall customer growth and retention trends.
- Renewal volumes stayed steady until 2022, then declined in 2023 due to pricing pressure and customer switching.
- New policy uptake has gradually fallen since 2021, reflecting weaker acquisition amid economic challenges.
- The drop in new policies aligns with the UK cost-of-living crisis and reduced household spending.
- Rising premiums and repair costs increased price sensitivity, prompting switching or reduced cover levels.
- Both policy types follow yearly cycles, peaking in high-risk months and dipping during lower-risk periods.
- Small, random residuals suggest the trends are driven by consistent seasonal and economic factors, not random shocks.

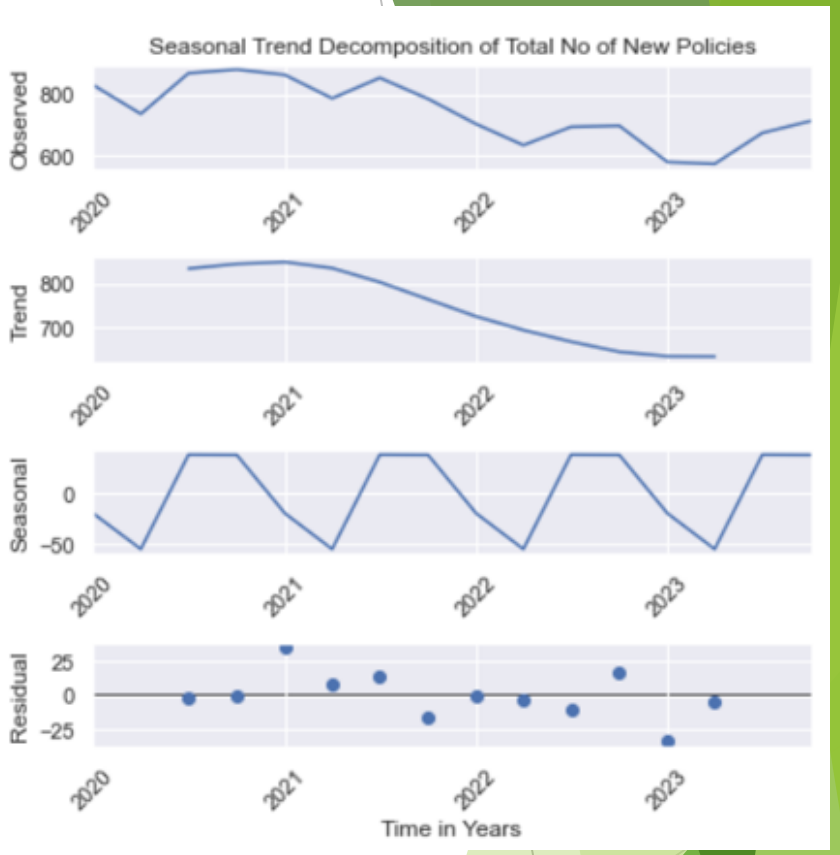

Figure 11.2- New Policy Data Decomposition

2. Pricing Level Base Analysis

- Renewals form a larger and more stable customer base.
- Average premiums are higher for renewals due to added cover or loyalty pricing.
- New policies are cheaper to attract customers with discounts.
- Gradual “price walking” increases renewal premiums over time.
- Renewal customers show higher value; new ones are more price-sensitive.

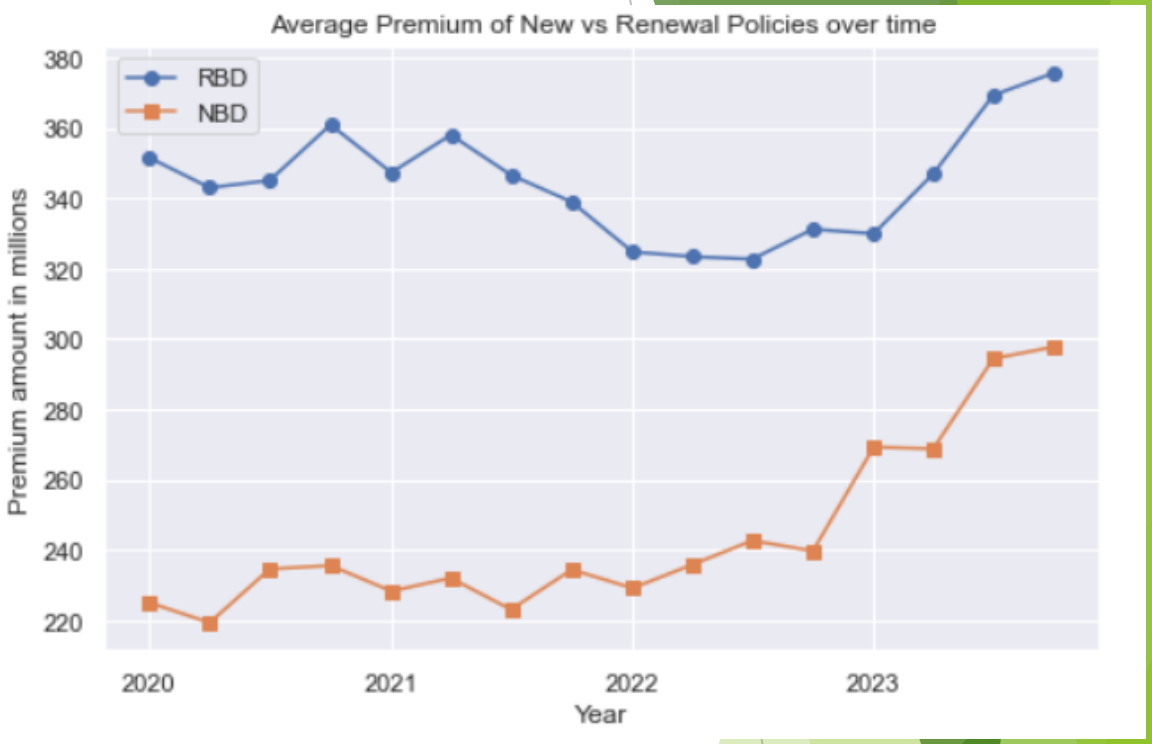

Figure 11.3 – Average Premium of New vs Renewal Policies

Revenue Contribution

The Figure 12.1 shows the quarterly contribution of both renewals and new business to the total premium amount. Following are the insights from the bar chart:

- Renewals remain the key revenue source, contributing most of the total written premiums each quarter from 2020–2023.
- Number of policies is declining for both new and renewal business, indicating slower customer growth and a possible risk to future revenue.
- Average premiums are increasing, likely as a pricing strategy to offset lower policy volumes and rising claim costs.
- Profitability is maintained, as higher premiums help balance the decline in policy count and keep margins stable.

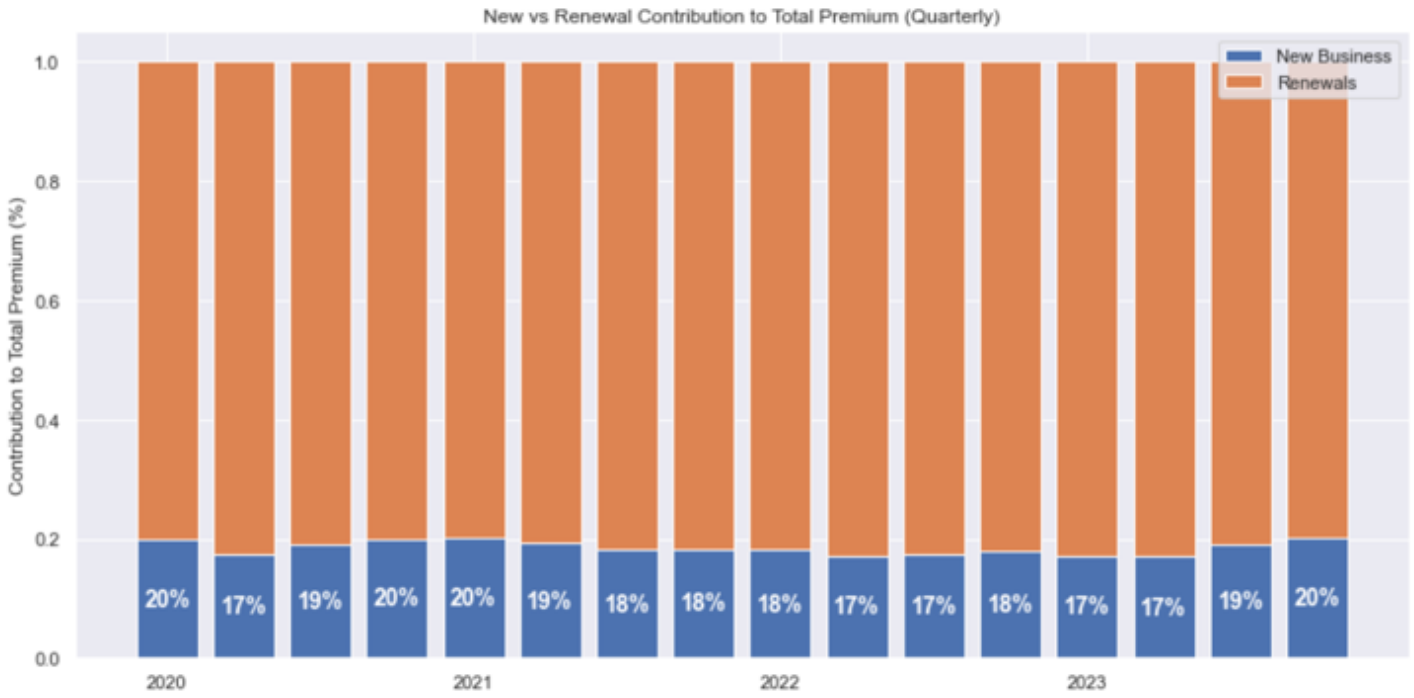

Figure 12.1- Bar chart showing quartelrley Revenue Contribution by both Renewals and New Business

Analysis of Impact of Pricing on Customer Acquisition and Retention

The correlation between average premium and number of policies indicates how pricing affects customer acquisition (new business) and retention (renewals).

Figure 13.1 shows scatter plot of new policies and *Figure 13.2* shows scatterplot of renewals

New policies show a moderate negative relationship (-0.56): higher average premiums are linked to fewer new customers. This suggests new customers are price-sensitive and more likely to shop around.

Renewals also show a negative relationship (-0.25), but it's weaker. Existing customers are less price-sensitive, often staying despite price increases due to convenience, loyalty, or inertia.

Some quarters deviate from the trend (dots away from the line), meaning factors like marketing campaigns, seasonality, or external events also influence policy counts beyond just price.

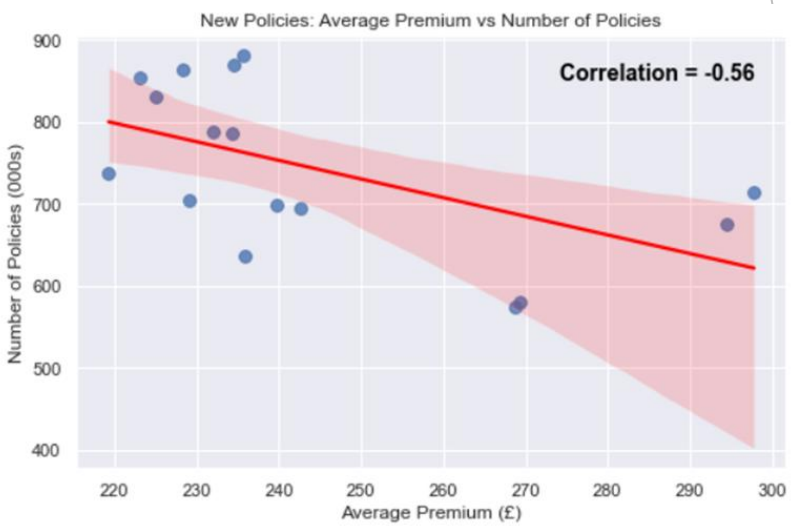

Figure 13.1 – Scatterplot of New Policies

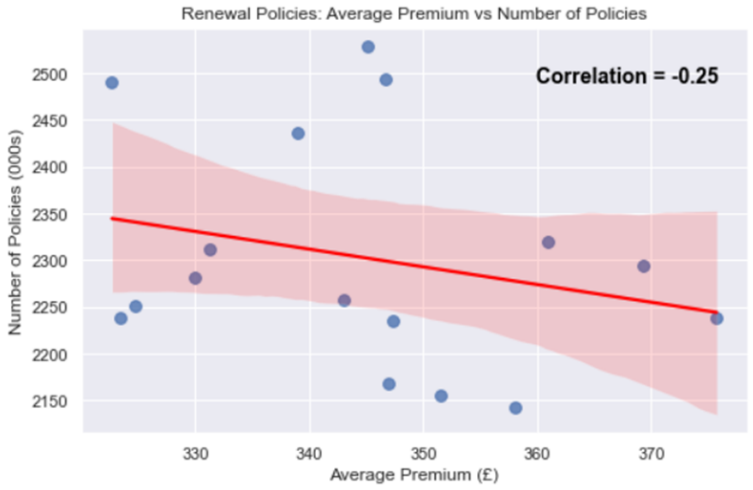

Figure 13.2 – Scatterplot of Renewal Policies

Conclusion

The overall analysis highlights that while the UK home insurance market has faced increased volatility due to inflation and extreme weather events, insurers have maintained profitability through effective pricing and underwriting adjustments.

- **Premium and Policy Trends:** Average premiums have risen steadily since 2021 to offset inflation and higher claim costs, while the number of policies has gradually declined, reflecting affordability pressures and greater price sensitivity among customers.

- **Claims Behaviour:** Claim frequency has fallen consistently, but claim severity and burn cost have increased, indicating fewer but more expensive claims driven by storm and freeze events such as those in 2014 and 2022.

- **Profitability and Risk Management:** Despite short-term spikes in loss ratio during high-claim periods, overall profitability remains stable due to timely premium adjustments and disciplined risk control.

- **Customer Dynamics:** Renewals remain the dominant revenue source, but both new and renewal policy counts are decreasing. New customers are more price-sensitive, whereas renewals show greater loyalty despite premium increases.

- **Market Drivers:** Inflation, supply-chain issues, and climate-related risks continue to shape pricing and profitability, highlighting the need for dynamic pricing models and improved catastrophe preparedness.

In short**,** the home insurance market shows strong long-term resilience. Strategic premium management and customer retention efforts have sustained profitability despite economic challenges and event-driven claim surges.

Future Scope

Using time series analysis, it was identified that both trend and seasonality exist in the data.
To build on this, a SARIMA forecasting model can be applied to predict key metrics such as:

- Number of claims
- Total claim amount
- Number of policies
- Average premium

From these forecasted values, future Burn Cost and Loss Ratio can be calculated to assess expected profitability.
This would help determine whether the projected average premium levels are sufficient to cover future risk or if pricing strategies need to be reviewed and adjusted to maintain profitability.

Thank You!!